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OPERATING SYSTEMS LABORATORY

Course Code: CSE309R01

Semester: V

Lab Manual

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Exercise No. 5. Simulation of CPU Scheduling with pre-emptive policies

Write programs to demonstrate pre-emptive scheduling algorithms and compare their performances

Objectives

Simulation of CPU Scheduling algorithm with pre-emption – Round robin,

SRTF Prerequisite

Knowledge of scheduling algorithms

Procedure

- ❑ Input the number of processes to be scheduled
- ❑ Input the arrival time, CPU burst time1, IO burst time and CPU burst time2 of each process
- ❑ Calculate the turnaround time and the waiting time of the processes based on the FCFS

Sample Input

Number of Processes = 4

Process	Arrival Time	CPU Burst 1	I/O Burst	CPU Burst 2
P1	0	4	2	3
P2	1	3	2	2
P3	2	2	1	2
P4	3	3	2	1

Time Quantum = 2

Output

```

Enter number of processes: 4

Enter Arrival Time, CPU Burst 1, IO Burst and CPU Burst 2:

P1 Arrival Time: 0
P1 CPU Burst 1: 4
P1 IO Burst: 2
P1 CPU Burst 2: 3

P2 Arrival Time: 1
P2 CPU Burst 1: 3
P2 IO Burst: 2
P2 CPU Burst 2: 2

P3 Arrival Time: 2
P3 CPU Burst 1: 2
P3 IO Burst: 1
P3 CPU Burst 2: 2

P4 Arrival Time: 3
P4 CPU Burst 1: 3
P4 IO Burst: 2
P4 CPU Burst 2: 1

Enter Time Quantum for Round Robin: 2

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===== ROUND ROBIN =====

```

```

Time Quantum = 2

```

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Gantt Chart:

```

```

| P1 | P1 | P2 | P2 | P3 | P2 | P1 | P3 | P4 | P4 | P3 | P3 | P3 | P3 |
P3 | P3 | P3 | P2 | P1 | P1 | P3 | P3 | P4 | P3 | P3 | P1 | P1 | P3 |
P3 | P1 | P1 | P2 | P4 | P1 | P1 | P2 | P2 | P3 | P3 | P1 | P1 | P2 |
P2 | P1 | P1 | P2 | P2 | P3 | P3 | P1 | P1 | P2 | P2 | P1 | P1 | P2 |
P2 | P1 | P1 | P2 | P2 | P1 | P1 | P1 | P1 | P2 | P2 | P4 |

```

Process	AT	CPU1	IO	CPU2	CT	TAT	WT	RT
P1	0	4	2	3	23	23	14	0
P2	1	3	2	2	33	32	25	1
P3	2	2	1	2	13	11	6	2
P4	3	3	2	1	68	65	59	5

```

Average Turnaround Time = 32.75
Average Waiting Time = 26.00
Average Response Time = 2.00
Throughput = 0.06 processes/unit time

```

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===== SRTF =====

Gantt Chart:
| P1 | P1 | P1 | P1 | P3 | P3 | P1 | P1 | P1 | P2 | P2 | P2 | P3 | P3 |
P2 | P2 | P4 | P4 | P4 | P4 |

Process AT  CPU1  I/O  CPU2  CT  TAT WT  RT
P1  0   4   2   3   9   9   0   0
P2  1   3   2   2  16  15   8   8
P3  2   2   1   2  14  12   7   2
P4  3   3   2   1  22  19  13  13

Average Turnaround Time = 13.75
Average Waiting Time = 7.00
Average Response Time = 5.75
Throughput = 0.18 processes/unit time
```

Observation

Workout a sample test case and estimate all times, Gantt chart for both round robin and SRTF scheduling policy with appropriate time quantum for round robin

A sample test case with four processes was considered using Arrival Time, CPU Burst 1, I/O Burst and CPU Burst 2. Round Robin scheduling was implemented with a time quantum of 2. SRTF scheduling was implemented by selecting the process with the shortest remaining CPU burst time. The Gantt charts, execution order, completion time, turnaround time, waiting time and response time were obtained for both scheduling algorithms.

Result

Thus, the pre-emptive CPU scheduling algorithms Round Robin and SRTF were successfully implemented. The Gantt charts and performance measures were obtained and compared for the given sample test case.

One thing: **use the Gantt chart exactly as your program printed it** if your faculty wants the actual output screenshot. The Round Robin output is printed one CPU time unit at a time, so it looks very long; that's coming from the program, not something you need to manually simplify.